Module-1

Ecosystem – Basic Concepts and Structure

1. An ecosystem consists of:
A) Only plants
B) Only animals
C) Living organisms and their physical environment
D) Only microorganisms
Answer: C) Living organisms and their physical environment

2. The living components of an ecosystem are called:
A) Abiotic components
B) Biotic components
C) Physical components
D) Climatic components
Answer: B) Biotic components

3. Which of the following is an abiotic component?
A) Plants
B) Animals
C) Sunlight
D) Bacteria
Answer: C) Sunlight

4. Which of the following is a biotic component?
A) Temperature
B) Water
C) Soil
D) Plants
Answer: D) Plants

5. Green plants in an ecosystem are called:
A) Producers
B) Consumers
C) Decomposers
D) Scavengers
Answer: A) Producers

6. Producers are also known as:
A) Autotrophs
B) Heterotrophs
C) Carnivores
D) Decomposers
Answer: A) Autotrophs

7. Organisms that depend on other organisms for food are called:
A) Producers
B) Consumers

C) Autotrophs
D) Abiotic components
Answer: B) Consumers

8. Organisms that feed directly on plants are called:
A) Primary consumers
B) Secondary consumers
C) Producers
D) Decomposers
Answer: A) Primary consumers

9. Herbivores primarily feed on:
A) Plants
B) Carnivores
C) Rocks
D) Soil
Answer: A) Plants

10. Carnivores primarily feed on:
A) Other animals
B) Plants only
C) Minerals
D) Sunlight
Answer: A) Other animals

11. Organisms that consume both plants and animals are called:
A) Omnivores
B) Producers
C) Autotrophs
D) Decomposers
Answer: A) Omnivores

12. Bacteria and fungi commonly function as:
A) Producers
B) Decomposers
C) Primary consumers
D) Abiotic factors
Answer: B) Decomposers

13. Decomposers primarily break down:
A) Dead organic matter
B) Sunlight
C) Rocks only
D) Water molecules
Answer: A) Dead organic matter

14. The primary source of energy for most ecosystems is:
A) Sun
B) Moon
C) Soil

D) Wind
Answer: A) Sun

15. The sequence of transfer of food energy from one organism to another is called a:
A) Food chain
B) Water cycle
C) Rock cycle
D) Nutrient tank
Answer: A) Food chain

16. A network of interconnected food chains is called a:
A) Food web
B) Food pyramid
C) Ecological niche
D) Habitat
Answer: A) Food web

17. Each feeding level in a food chain is known as a:
A) Trophic level
B) Water level
C) Pollution level
D) Ground level
Answer: A) Trophic level

18. The first trophic level generally consists of:
A) Producers
B) Herbivores
C) Carnivores
D) Decomposers
Answer: A) Producers

19. The second trophic level generally consists of:
A) Primary consumers
B) Producers
C) Abiotic components
D) Minerals
Answer: A) Primary consumers

20. The graphical representation of trophic levels is called an:
A) Ecological pyramid
B) Water pyramid
C) Energy cycle
D) Ecological map
Answer: A) Ecological pyramid

21. The pyramid showing the amount of energy at different trophic levels is called:
A) Pyramid of energy
B) Pyramid of water
C) Pyramid of rainfall

D) Pyramid of soil
Answer: A) Pyramid of energy

22. Energy flow in an ecosystem is generally:
A) Unidirectional
B) Completely cyclic
C) Absent
D) Reversible at every stage
Answer: A) Unidirectional

23. The natural place where an organism lives is called its:
A) Habitat
B) Trophic level
C) Food web
D) Population
Answer: A) Habitat

24. The functional role of an organism in an ecosystem is called its:
A) Ecological niche
B) Habitat
C) Population
D) Community
Answer: A) Ecological niche

25. A group of organisms of the same species living in an area is called a:
A) Population
B) Community
C) Ecosystem
D) Biosphere
Answer: A) Population

26. An ecosystem dominated by trees is called a:
A) Forest ecosystem
B) Desert ecosystem
C) Riverine ecosystem
D) Oceanic ecosystem
Answer: A) Forest ecosystem

27. The major producers in a forest ecosystem are:
A) Trees and green plants
B) Carnivores
C) Decomposers only
D) Rocks
Answer: A) Trees and green plants

28. Which of the following is a primary consumer in a forest ecosystem?
A) Deer
B) Tiger
C) Eagle

D) Fungi
Answer: A) Deer

29. Which of the following can act as a carnivore in a forest ecosystem?
A) Tiger
B) Grass
C) Tree
D) Shrub
Answer: A) Tiger

30. Forests are important reservoirs of:
A) Biodiversity
B) Industrial waste
C) Sewage
D) Fossil fuel only
Answer: A) Biodiversity

31. Forest vegetation helps prevent:
A) Soil erosion
B) Photosynthesis
C) Nutrient cycling
D) Biodiversity
Answer: A) Soil erosion

32. Forests help regulate atmospheric carbon dioxide through:
A) Photosynthesis
B) Combustion
C) Incineration
D) Landfilling
Answer: A) Photosynthesis

33. Large-scale removal of forest vegetation is called:
A) Deforestation
B) Afforestation
C) Reforestation
D) Conservation
Answer: A) Deforestation

34. Establishing trees on land to create forest cover is called:
A) Afforestation
B) Deforestation
C) Desertification
D) Eutrophication
Answer: A) Afforestation

35. Forests provide habitat for:
A) Diverse plants and animals
B) Only trees
C) Only microorganisms

D) Only humans
Answer: A) Diverse plants and animals

36. Desert ecosystems are characterized by:
A) Low rainfall
B) Continuous heavy rainfall
C) Permanent flooding
D) Abundant surface water
Answer: A) Low rainfall

37. Plants adapted to dry environments are called:
A) Xerophytes
B) Hydrophytes
C) Phytoplankton
D) Zooplankton
Answer: A) Xerophytes

38. Which is a typical desert plant?
A) Cactus
B) Lotus
C) Water lily
D) Mangrove
Answer: A) Cactus

39. A major limiting factor in a desert ecosystem is:
A) Water
B) Sand
C) Sunlight
D) Air
Answer: A) Water

40. Desert vegetation is generally:
A) Sparse
B) Very dense everywhere
C) Completely aquatic
D) Dominated by phytoplankton
Answer: A) Sparse

41. Desert plants commonly possess adaptations for:
A) Water conservation
B) Increasing water loss
C) Permanent flooding
D) Aquatic respiration
Answer: A) Water conservation

42. Animals living in deserts are adapted mainly to:
A) Dry conditions
B) Permanent flooding
C) Deep ocean conditions

D) Polar conditions only
Answer: A) Dry conditions

43. Wetlands are areas where land is:
A) Saturated or covered with water for significant periods
B) Permanently dry
C) Always covered by sand
D) Completely devoid of water
Answer: A) Saturated or covered with water for significant periods

44. Which of the following is a wetland?
A) Marsh
B) Desert
C) Mountain peak
D) Sand dune
Answer: A) Marsh

45. A swamp is an example of a:
A) Wetland
B) Desert
C) Ocean
D) Grassland only
Answer: A) Wetland

46. Plants adapted to waterlogged conditions are called:
A) Hydrophytes
B) Xerophytes
C) Carnivores
D) Decomposers
Answer: A) Hydrophytes

47. Wetlands can help in:
A) Flood regulation
B) Increasing desertification
C) Increasing erosion
D) Eliminating biodiversity
Answer: A) Flood regulation

48. Wetlands can function as natural:
A) Water filters
B) Noise generators
C) Fossil-fuel plants
D) Incinerators
Answer: A) Water filters

49. Wetlands provide important habitat for:
A) Aquatic and semi-aquatic organisms
B) Only desert organisms
C) Only forest trees

D) Only humans
Answer: A) Aquatic and semi-aquatic organisms

50. An ecosystem associated with rivers and streams is called:
A) Riverine ecosystem
B) Desert ecosystem
C) Oceanic ecosystem
D) Forest ecosystem
Answer: A) Riverine ecosystem

51. Riverine ecosystems are mainly characterized by:
A) Flowing freshwater
B) Saline ocean water
C) Dry sand
D) Permanent drought
Answer: A) Flowing freshwater

52. Vegetation growing along riverbanks is called:
A) Riparian vegetation
B) Desert vegetation
C) Marine vegetation
D) Alpine vegetation
Answer: A) Riparian vegetation

53. Which organism is commonly associated with riverine ecosystems?
A) Fish
B) Cactus
C) Camel
D) Desert shrub
Answer: A) Fish

54. Pollution of rivers can directly affect:
A) Aquatic organisms
B) Desert vegetation only
C) Mountain rocks only
D) Solar radiation
Answer: A) Aquatic organisms

55. Riverine ecosystems are generally classified as:
A) Freshwater ecosystems
B) Marine ecosystems
C) Desert ecosystems
D) Terrestrial forests
Answer: A) Freshwater ecosystems

56. Oceanic ecosystems are characterized by:
A) Saline water
B) Freshwater only
C) Dry soil

D) Low atmospheric pressure only
Answer: A) Saline water

57. Microscopic photosynthetic organisms in oceans are called:
A) Phytoplankton
B) Zooplankton
C) Carnivores
D) Scavengers
Answer: A) Phytoplankton

58. Phytoplankton function mainly as:
A) Producers
B) Secondary consumers
C) Tertiary consumers
D) Decomposers only
Answer: A) Producers

59. Zooplankton generally function as:
A) Consumers
B) Producers
C) Abiotic factors
D) Minerals
Answer: A) Consumers

60. Coral reefs are part of:
A) Marine ecosystems
B) Desert ecosystems
C) Forest ecosystems
D) Riverine ecosystems only
Answer: A) Marine ecosystems

61. Oceans cover approximately _____ of the Earth's surface.
A) 71%
B) 20%
C) 35%
D) 10%
Answer: A) 71%

62. Ocean ecosystems play an important role in regulating:
A) Global climate
B) Road traffic
C) Building construction
D) Noise levels only
Answer: A) Global climate

63. The major producers forming the base of many marine food chains are:
A) Phytoplankton
B) Sharks
C) Large fish

D) Whales
Answer: A) Phytoplankton

64. Sustainability refers to meeting present needs while considering:
A) Future generations
B) Only present consumption
C) Only industrial growth
D) Only population growth
Answer: A) Future generations

65. Sustainable development integrates:
A) Environmental, social and economic dimensions
B) Only economic development
C) Only industrial development
D) Only environmental protection
Answer: A) Environmental, social and economic dimensions

66. SDG stands for:
A) Sustainable Development Goal
B) Social Development Group
C) Sustainable Design Guide
D) Social Development Goalpost
Answer: A) Sustainable Development Goal

67. The total number of Sustainable Development Goals is:
A) 17
B) 15
C) 20
D) 25
Answer: A) 17

68. The SDGs form part of the:
A) 2030 Agenda for Sustainable Development
B) 2050 Energy Plan only
C) 2020 Water Plan
D) 2040 Forest Plan
Answer: A) 2030 Agenda for Sustainable Development

69. The target year associated with the 2030 Agenda is:
A) 2030
B) 2040
C) 2050
D) 2025
Answer: A) 2030

70. SDG 1 is:
A) No Poverty
B) Zero Hunger
C) Quality Education

D) Climate Action
Answer: A) No Poverty

71. SDG 2 is:
A) Zero Hunger
B) Good Health and Well-being
C) Gender Equality
D) Life on Land
Answer: A) Zero Hunger

72. SDG 3 is:
A) Good Health and Well-being
B) Quality Education
C) No Poverty
D) Climate Action
Answer: A) Good Health and Well-being

73. SDG 4 is:
A) Quality Education
B) Gender Equality
C) Clean Water and Sanitation
D) Affordable and Clean Energy
Answer: A) Quality Education

74. SDG 5 is:
A) Gender Equality
B) Zero Hunger
C) Climate Action
D) Life Below Water
Answer: A) Gender Equality

75. SDG 6 is:
A) Clean Water and Sanitation
B) Affordable and Clean Energy
C) Climate Action
D) Life on Land
Answer: A) Clean Water and Sanitation

76. SDG 7 is:
A) Affordable and Clean Energy
B) Clean Water and Sanitation
C) Sustainable Cities and Communities
D) Life Below Water
Answer: A) Affordable and Clean Energy

77. SDG 8 is:
A) Decent Work and Economic Growth
B) Reduced Inequalities
C) Climate Action

D) Zero Hunger
Answer: A) Decent Work and Economic Growth

78. SDG 9 is:
A) Industry, Innovation and Infrastructure
B) Quality Education
C) Life on Land
D) Gender Equality
Answer: A) Industry, Innovation and Infrastructure

79. SDG 10 is:
A) Reduced Inequalities
B) No Poverty
C) Climate Action
D) Life Below Water
Answer: A) Reduced Inequalities

80. SDG 11 is:
A) Sustainable Cities and Communities
B) Climate Action
C) Zero Hunger
D) Gender Equality
Answer: A) Sustainable Cities and Communities

81. SDG 12 is:
A) Responsible Consumption and Production
B) Quality Education
C) Clean Water and Sanitation
D) No Poverty
Answer: A) Responsible Consumption and Production

82. SDG 13 is:
A) Climate Action
B) Life Below Water
C) Life on Land
D) Zero Hunger
Answer: A) Climate Action

83. SDG 14 is:
A) Life Below Water
B) Life on Land
C) Climate Action
D) No Poverty
Answer: A) Life Below Water

84. SDG 15 is:
A) Life on Land
B) Life Below Water
C) Clean Water and Sanitation

D) Sustainable Cities and Communities
Answer: A) Life on Land

85. SDG 16 is:
A) Peace, Justice and Strong Institutions
B) Partnerships for the Goals
C) Climate Action
D) Quality Education
Answer: A) Peace, Justice and Strong Institutions

86. SDG 17 is:
A) Partnerships for the Goals
B) No Poverty
C) Zero Hunger
D) Climate Action
Answer: A) Partnerships for the Goals

87. Providing access to safe drinking water supports:
A) SDG 6 – Clean Water and Sanitation
B) SDG 4 – Quality Education
C) SDG 9 – Industry, Innovation and Infrastructure
D) SDG 16 – Peace, Justice and Strong Institutions
Answer: A) SDG 6 – Clean Water and Sanitation

88. Promoting solar and wind power supports:
A) SDG 7 – Affordable and Clean Energy
B) SDG 2 – Zero Hunger
C) SDG 5 – Gender Equality
D) SDG 16 – Peace, Justice and Strong Institutions
Answer: A) SDG 7 – Affordable and Clean Energy

89. Developing safe and resilient urban areas supports:
A) SDG 11 – Sustainable Cities and Communities
B) SDG 2 – Zero Hunger
C) SDG 4 – Quality Education
D) SDG 14 – Life Below Water
Answer: A) SDG 11 – Sustainable Cities and Communities

90. Practising reduce, reuse and recycle primarily supports:
A) SDG 12 – Responsible Consumption and Production
B) SDG 5 – Gender Equality
C) SDG 4 – Quality Education
D) SDG 16 – Peace, Justice and Strong Institutions
Answer: A) SDG 12 – Responsible Consumption and Production

91. Reducing greenhouse-gas emissions is an action associated with:
A) SDG 13 – Climate Action
B) SDG 1 – No Poverty
C) SDG 4 – Quality Education

D) SDG 10 – Reduced Inequalities
Answer: A) SDG 13 – Climate Action

92. Protecting oceans and marine ecosystems supports:
A) SDG 14 – Life Below Water
B) SDG 15 – Life on Land
C) SDG 8 – Decent Work and Economic Growth
D) SDG 9 – Industry, Innovation and Infrastructure
Answer: A) SDG 14 – Life Below Water

93. Protecting forests and terrestrial biodiversity supports:
A) SDG 15 – Life on Land
B) SDG 14 – Life Below Water
C) SDG 7 – Affordable and Clean Energy
D) SDG 10 – Reduced Inequalities
Answer: A) SDG 15 – Life on Land

94. Preventing food wastage is an action associated particularly with:
A) Responsible Consumption and Production
B) Increased resource consumption
C) Increased waste generation
D) Deforestation
Answer: A) Responsible Consumption and Production

95. Using public transport instead of individual vehicles can support:
A) Sustainable Cities and Communities
B) Increased traffic emissions
C) Increased fossil-fuel consumption
D) Increased urban pollution
Answer: A) Sustainable Cities and Communities

96. Switching off lights and appliances when not required is an example of:
A) Energy conservation
B) Energy wastage
C) Water pollution
D) Deforestation
Answer: A) Energy conservation

97. Planting and protecting trees is an action that supports:
A) Life on Land
B) Life Below Water only
C) Gender Equality only
D) Quality Education only
Answer: A) Life on Land

98. Avoiding unnecessary single-use products is an example of:
A) Responsible consumption
B) Resource wastage
C) Open dumping

D) Unsustainable production
Answer: A) Responsible consumption

99. Rainwater harvesting is an action that contributes to:
A) Water conservation
B) Water wastage
C) Air pollution
D) Noise pollution
Answer: A) Water conservation

100. Cooperation among governments, institutions and communities to achieve sustainable development is reflected in:
A) SDG 17 – Partnerships for the Goals
B) SDG 2 – Zero Hunger
C) SDG 5 – Gender Equality
D) SDG 14 – Life Below Water
Answer: A) SDG 17 – Partnerships for the Goals

Module-2

Water availability

1. The largest reservoir of water on Earth is:
A) Rivers
B) Lakes
C) Oceans
D) Groundwater
Answer: C) Oceans

2. The major natural source of freshwater replenishment is:
A) Rainfall
B) Seawater
C) Industrial wastewater
D) Domestic sewage
Answer: A) Rainfall

3. Water present below the Earth's surface is called:
A) Surface water
B) Groundwater
C) Seawater
D) Atmospheric water
Answer: B) Groundwater

4. Which of the following is a surface-water source?
A) Borewell
B) Tube well
C) River
D) Aquifer
Answer: C) River

5. Which of the following is a groundwater source?
A) River
B) Lake
C) Well
D) Reservoir
Answer: C) Well

6. A geological formation that stores and transmits groundwater is called:
A) Aquifer
B) Estuary
C) Lagoon
D) Watershed
Answer: A) Aquifer

7. The upper surface of groundwater in an unconfined aquifer is called the:
A) Water table
B) Water divide

C) Watershed
D) Aqueduct
Answer: A) Water table

8. The process by which water enters the soil surface is called:
A) Infiltration
B) Evaporation
C) Condensation
D) Precipitation
Answer: A) Infiltration

9. The downward movement of water through soil and porous materials is called:
A) Percolation
B) Evaporation
C) Condensation
D) Sublimation
Answer: A) Percolation

10. Rainwater harvesting is mainly used for:
A) Water conservation
B) Increasing water pollution
C) Increasing runoff
D) Increasing water wastage
Answer: A) Water conservation

11. Groundwater recharge can be increased through:
A) Rainwater harvesting
B) Deforestation
C) Over-pumping
D) Paving all open land
Answer: A) Rainwater harvesting

12. A structure constructed mainly for storing surface water is a:
A) Reservoir
B) Aquifer
C) Borewell
D) Tube well
Answer: A) Reservoir

13. Excessive extraction of groundwater can cause:
A) Decline in the water table
B) Increase in rainfall
C) Increase in river flow
D) Increase in seawater volume
Answer: A) Decline in the water table

14. Sustainable water management emphasizes:
A) Conservation and efficient use
B) Uncontrolled withdrawal
C) Water wastage

D) Pollution of water bodies
Answer: A) Conservation and efficient use

15. Water scarcity refers to a condition in which:
A) Available usable water is insufficient to meet demand
B) Water is always abundant
C) Rainfall occurs every day
D) Groundwater is never used
Answer: A) Available usable water is insufficient to meet demand

16. Water quality is commonly assessed using:
A) Physical, chemical and biological characteristics
B) Economic characteristics only
C) Geological characteristics only
D) Climatic characteristics only
Answer: A) Physical, chemical and biological characteristics

17. Which of the following is a physical water-quality parameter?
A) Turbidity
B) Fluoride
C) Chloride
D) Nitrate
Answer: A) Turbidity

18. The acidity or alkalinity of water is expressed by:
A) pH
B) BOD
C) COD
D) DO
Answer: A) pH

19. A pH value of 7 indicates:
A) Neutral condition
B) Acidic condition
C) Strongly alkaline condition
D) Highly saline condition
Answer: A) Neutral condition

20. The cloudiness of water caused by suspended particles is called:
A) Turbidity
B) Hardness
C) Acidity
D) Alkalinity
Answer: A) Turbidity

21. Water hardness is mainly caused by the presence of salts of:
A) Calcium and magnesium
B) Sodium and potassium only
C) Hydrogen and oxygen

D) Carbon and nitrogen
Answer: A) Calcium and magnesium

22. TDS stands for:
A) Total Dissolved Solids
B) Total Domestic Sewage
C) Total Disinfection System
D) Treated Dissolved Sewage
Answer: A) Total Dissolved Solids

23. The presence of disease-causing microorganisms in water represents:
A) Biological contamination
B) Noise pollution
C) Thermal pollution
D) Light pollution
Answer: A) Biological contamination

24. Drinking water should be free from:
A) Pathogenic microorganisms
B) Dissolved oxygen
C) All minerals
D) Water molecules
Answer: A) Pathogenic microorganisms

25. The main purpose of drinking-water disinfection is to:
A) Inactivate disease-causing microorganisms
B) Increase turbidity
C) Increase hardness
D) Increase suspended solids
Answer: A) Inactivate disease-causing microorganisms

26. Diseases transmitted through contaminated drinking water are called:
A) Water-borne diseases
B) Genetic diseases
C) Occupational diseases
D) Air-borne diseases
Answer: A) Water-borne diseases

27. Which of the following is a water-borne disease?
A) Cholera
B) Asthma
C) Diabetes
D) Hypertension
Answer: A) Cholera

28. Cholera is caused by:
A) Vibrio cholerae
B) Plasmodium
C) Influenza virus

D) Aspergillus
Answer: A) Vibrio cholerae

29. Typhoid is caused by:
A) Salmonella Typhi
B) Vibrio cholerae
C) Plasmodium vivax
D) Dengue virus
Answer: A) Salmonella Typhi

30. Typhoid can spread through:
A) Contaminated food and water
B) Loud noise
C) Sunlight
D) Wind
Answer: A) Contaminated food and water

31. Which disease is commonly associated with contaminated water and poor sanitation?
A) Dysentery
B) Diabetes
C) Hypertension
D) Hearing loss
Answer: A) Dysentery

32. Hepatitis A may spread through:
A) Contaminated food and water
B) Noise pollution
C) Wind energy
D) Solar radiation
Answer: A) Contaminated food and water

33. Poor sanitation can increase the occurrence of:
A) Water-borne diseases
B) Noise pollution
C) Wind erosion only
D) Solar radiation
Answer: A) Water-borne diseases

34. Boiling water before drinking primarily helps reduce:
A) Microbial contamination
B) Water hardness completely
C) Fluoride concentration
D) Salinity completely
Answer: A) Microbial contamination

35. Chlorination is commonly used for:
A) Disinfection of water
B) Increasing hardness
C) Increasing turbidity

D) Increasing salinity
Answer: A) Disinfection of water

36. Diseases associated with the chemical quality or other water-related conditions may be referred to as:
A) Water-induced diseases
B) Genetic diseases only
C) Occupational diseases
D) Noise-induced diseases
Answer: A) Water-induced diseases

37. Fluorosis is associated with excessive intake of:
A) Fluoride
B) Iron
C) Oxygen
D) Nitrogen
Answer: A) Fluoride

38. Providing safe drinking water is an important measure for preventing:
A) Water-related diseases
B) Noise pollution
C) Soil erosion
D) Wind erosion
Answer: A) Water-related diseases

39. Excessive fluoride in drinking water can cause:
A) Fluorosis
B) Cholera
C) Typhoid
D) Malaria
Answer: A) Fluorosis

40. Fluorosis affecting the teeth is called:
A) Dental fluorosis
B) Skeletal fluorosis
C) Acute fluorosis only
D) Typhoid
Answer: A) Dental fluorosis

41. Fluorosis affecting bones and joints is called:
A) Skeletal fluorosis
B) Dental fluorosis
C) Cholera
D) Dysentery
Answer: A) Skeletal fluorosis

42. Long-term consumption of water containing excessive fluoride mainly affects:
A) Teeth and bones
B) Ears only
C) Hair only

D) Eyes only
Answer: A) Teeth and bones

43. A visible sign associated with dental fluorosis is:
A) Mottling or discoloration of teeth
B) Hearing loss
C) Hair loss
D) Fever
Answer: A) Mottling or discoloration of teeth

44. The process of removing excess fluoride from drinking water is called:
A) Defluoridation
B) Fluoridation
C) Sedimentation
D) Screening
Answer: A) Defluoridation

45. The Nalgonda technique is associated with:
A) Defluoridation of water
B) Air-pollution control
C) Noise control
D) Sewage screening
Answer: A) Defluoridation of water

46. Fluoride contamination is an important concern in some:
A) Groundwater sources
B) Noise barriers
C) Solar panels
D) Wind turbines
Answer: A) Groundwater sources

47. A suitable measure in a fluoride-affected area is:
A) Providing drinking water with acceptable fluoride concentration
B) Consuming untreated high-fluoride water
C) Increasing fluoride concentration
D) Mixing drinking water with wastewater
Answer: A) Providing drinking water with acceptable fluoride concentration

48. Fluoride in drinking water becomes a health concern particularly when its concentration is:
A) Excessive
B) Always zero
C) Unrelated to exposure
D) Equal to dissolved oxygen
Answer: A) Excessive

49. The capacity to do work is called:
A) Energy
B) Pressure
C) Density

D) Velocity
Answer: A) Energy

50. The SI unit of energy is:
A) Joule
B) Watt
C) Newton
D) Pascal
Answer: A) Joule

51. Energy possessed by an object due to its motion is:
A) Kinetic energy
B) Potential energy
C) Chemical energy
D) Nuclear energy
Answer: A) Kinetic energy

52. Energy possessed by an object due to its position is:
A) Potential energy
B) Kinetic energy
C) Sound energy
D) Wind energy
Answer: A) Potential energy

53. Energy stored in chemical bonds is called:
A) Chemical energy
B) Wind energy
C) Tidal energy
D) Sound energy
Answer: A) Chemical energy

54. Heat energy is also known as:
A) Thermal energy
B) Potential energy
C) Nuclear energy
D) Tidal energy
Answer: A) Thermal energy

55. Energy obtained from atomic nuclei is called:
A) Nuclear energy
B) Wind energy
C) Solar energy
D) Tidal energy
Answer: A) Nuclear energy

56. Which of the following is a conventional energy source?
A) Coal
B) Solar energy
C) Wind energy

D) Tidal energy
Answer: A) Coal

57. Which of the following is a fossil fuel?
A) Petroleum
B) Solar radiation
C) Wind
D) Tidal energy
Answer: A) Petroleum

58. Natural gas is classified as a:
A) Fossil fuel
B) Solar-energy source
C) Wind-energy source
D) Tidal-energy source
Answer: A) Fossil fuel

59. Coal is a:
A) Non-renewable energy resource
B) Renewable energy resource
C) Solar-energy resource
D) Wind-energy resource
Answer: A) Non-renewable energy resource

60. Fossil fuels are considered non-renewable because they:
A) Require very long geological periods to form
B) Are produced every day
C) Are unlimited
D) Are produced by sunlight instantly
Answer: A) Require very long geological periods to form

61. Burning fossil fuels releases significant amounts of:
A) Carbon dioxide
B) Hydrogen only
C) Helium only
D) Neon only
Answer: A) Carbon dioxide

62. Which fuel is commonly used in coal-based thermal power plants?
A) Coal
B) Wind
C) Sunlight
D) Tidal water
Answer: A) Coal

63. Petrol and diesel are obtained mainly from:
A) Petroleum
B) Wind
C) Solar energy

D) Biomass only
Answer: A) Petroleum

64. Sources of energy that are naturally replenished are called:
A) Renewable energy sources
B) Fossil fuels
C) Exhaustible resources
D) Mineral fuels
Answer: A) Renewable energy sources

65. Which of the following is a non-conventional energy source?
A) Solar energy
B) Coal
C) Petroleum
D) Diesel
Answer: A) Solar energy

66. Which of the following is a renewable energy source?
A) Wind energy
B) Coal
C) Petroleum
D) Natural gas
Answer: A) Wind energy

67. Energy obtained from organic materials is called:
A) Biomass energy
B) Nuclear energy
C) Petroleum energy
D) Coal energy
Answer: A) Biomass energy

68. Energy obtained from heat within the Earth is called:
A) Geothermal energy
B) Wind energy
C) Solar energy
D) Chemical energy
Answer: A) Geothermal energy

69. Energy obtained from the periodic rise and fall of seawater is called:
A) Tidal energy
B) Solar energy
C) Coal energy
D) Petroleum energy
Answer: A) Tidal energy

70. Biogas is produced mainly through:
A) Anaerobic decomposition of organic matter
B) Burning coal
C) Petroleum refining

D) Nuclear fission
Answer: A) Anaerobic decomposition of organic matter

71. The major combustible component of biogas is:
A) Methane
B) Oxygen
C) Nitrogen
D) Helium
Answer: A) Methane

72. Renewable energy can help reduce dependence on:
A) Fossil fuels
B) Sunlight
C) Wind
D) Biomass
Answer: A) Fossil fuels

73. The source of solar energy is the:
A) Sun
B) Moon
C) Ocean
D) Soil
Answer: A) Sun

74. Solar energy is classified as:
A) Renewable energy
B) Non-renewable energy
C) Fossil energy
D) Nuclear energy
Answer: A) Renewable energy

75. A photovoltaic cell converts solar energy directly into:
A) Electrical energy
B) Wind energy
C) Nuclear energy
D) Sound energy
Answer: A) Electrical energy

76. PV stands for:
A) Photovoltaic
B) Potential Voltage
C) Power Variation
D) Pressure Value
Answer: A) Photovoltaic

77. A solar water heater converts solar radiation mainly into:
A) Thermal energy
B) Nuclear energy
C) Wind energy

D) Sound energy
Answer: A) Thermal energy

78. A device designed to capture solar thermal energy is called a:
A) Solar collector
B) Wind turbine
C) Fuel cell
D) Generator only
Answer: A) Solar collector

79. A solar cooker uses:
A) Solar energy
B) Wind energy
C) Tidal energy
D) Nuclear energy
Answer: A) Solar energy

80. Solar photovoltaic systems generate electricity without direct combustion of:
A) Fossil fuels
B) Sunlight
C) Silicon
D) Electricity
Answer: A) Fossil fuels

81. Solar-energy availability is directly influenced by:
A) Solar radiation
B) Water hardness
C) Soil pH
D) Noise level
Answer: A) Solar radiation

82. Solar panels are commonly installed on rooftops to generate:
A) Electricity
B) Coal
C) Petroleum
D) Natural gas
Answer: A) Electricity

83. Wind energy is obtained from:
A) Moving air
B) Coal combustion
C) Petroleum refining
D) Groundwater
Answer: A) Moving air

84. A wind turbine converts the kinetic energy of wind mainly into:
A) Electrical energy
B) Nuclear energy
C) Chemical energy

D) Geothermal energy
Answer: A) Electrical energy

85. A group of wind turbines installed in one location is called a:
A) Wind farm
B) Solar farm
C) Thermal station
D) Reservoir
Answer: A) Wind farm

86. The blades of a wind turbine are attached to the:
A) Rotor
B) Boiler
C) Condenser
D) Solar collector
Answer: A) Rotor

87. Wind-energy generation depends strongly on:
A) Wind speed
B) Water hardness
C) Water pH
D) Turbidity
Answer: A) Wind speed

88. The instrument used to measure wind speed is:
A) Anemometer
B) Barometer
C) Thermometer
D) Hygrometer
Answer: A) Anemometer

89. Wind is mainly produced due to:
A) Uneven heating of the Earth's surface
B) Water chlorination
C) Waste recycling
D) Groundwater pumping
Answer: A) Uneven heating of the Earth's surface

90. Wind energy is a:
A) Renewable energy source
B) Fossil fuel
C) Non-renewable source
D) Petroleum product
Answer: A) Renewable energy source

91. Hydrogen is considered an alternative:
A) Energy carrier
B) Fossil-fuel deposit
C) Radioactive mineral

D) Solid waste
Answer: A) Energy carrier

92. A device that electrochemically converts the chemical energy of hydrogen into electricity is called a:
A) Fuel cell
B) Wind turbine
C) Solar collector
D) Boiler
Answer: A) Fuel cell

93. The main reaction product of a hydrogen fuel cell using hydrogen and oxygen is:
A) Water
B) Coal
C) Methane
D) Sulfur dioxide
Answer: A) Water

94. Hydrogen produced using renewable electricity through water electrolysis is called:
A) Green hydrogen
B) Brown coal
C) Biogas
D) Natural gas
Answer: A) Green hydrogen

95. The process of splitting water into hydrogen and oxygen using electricity is called:
A) Electrolysis
B) Filtration
C) Sedimentation
D) Chlorination
Answer: A) Electrolysis

96. Hydrogen can be used to generate electricity in a:
A) Fuel cell
B) Sedimentation tank
C) Clarifier
D) Cyclone separator
Answer: A) Fuel cell

97. Hydrogen fuel-cell vehicles use hydrogen as an:
A) Energy carrier
B) Air pollutant
C) Solid waste
D) Water contaminant
Answer: A) Energy carrier

98. Green hydrogen production is associated with electricity obtained from:
A) Renewable energy sources
B) Coal only
C) Diesel only

D) Petroleum only
Answer: A) Renewable energy sources

99. Which renewable source can provide electricity for green hydrogen production?
A) Solar energy
B) Coal
C) Diesel
D) Petrol
Answer: A) Solar energy

100. Solar energy, wind energy and green hydrogen can contribute to:
A) Reduced dependence on fossil fuels
B) Increased dependence on coal
C) Increased petroleum consumption
D) Depletion of renewable resources
Answer: A) Reduced dependence on fossil fuels

Module-3

1. Water pollution refers to:
A) Contamination of water that adversely affects its quality
B) Increase in rainfall
C) Groundwater recharge
D) Water conservation
Answer: A) Contamination of water that adversely affects its quality

2. Which of the following is a major source of water pollution?
A) Domestic sewage
B) Solar radiation
C) Wind energy
D) Rainwater harvesting
Answer: A) Domestic sewage

3. Wastewater generated from households is commonly called:
A) Domestic sewage
B) Industrial effluent
C) Storm water only
D) Groundwater
Answer: A) Domestic sewage

4. Wastewater discharged from industries is called:
A) Industrial effluent
B) Drinking water
C) Rainwater
D) Groundwater
Answer: A) Industrial effluent

5. Agricultural runoff may contain:
A) Fertilizers and pesticides
B) Only oxygen
C) Only sand
D) Only drinking water
Answer: A) Fertilizers and pesticides

6. Excessive nutrients entering a water body may cause:
A) Eutrophication
B) Afforestation
C) Desertification
D) Reforestation
Answer: A) Eutrophication

7. The major nutrients associated with eutrophication are:
A) Nitrogen and phosphorus
B) Hydrogen and helium
C) Gold and silver
D) Iron and copper only
Answer: A) Nitrogen and phosphorus

8. Excessive growth of algae in nutrient-rich water is called:
A) Algal bloom
B) Afforestation
C) Desertification
D) Sedimentation
Answer: A) Algal bloom

9. BOD stands for:
A) Biochemical Oxygen Demand
B) Biological Organic Density
C) Basic Oxygen Demand
D) Biochemical Oxygen Density
Answer: A) Biochemical Oxygen Demand

10. High BOD in water generally indicates:
A) High biodegradable organic pollution
B) Very pure water
C) Absence of microorganisms
D) High drinking-water quality
Answer: A) High biodegradable organic pollution

11. DO stands for:
A) Dissolved Oxygen
B) Domestic Oxygen
C) Dissolved Organic matter
D) Density of Oxygen
Answer: A) Dissolved Oxygen

12. Low dissolved oxygen in water can adversely affect:
A) Aquatic organisms
B) Solar panels
C) Wind turbines
D) Buildings
Answer: A) Aquatic organisms

13. COD stands for:
A) Chemical Oxygen Demand
B) Carbon Oxygen Density
C) Chemical Organic Density
D) Common Oxygen Demand
Answer: A) Chemical Oxygen Demand

14. Which pollutant can increase the turbidity of water?
A) Suspended solids
B) Dissolved oxygen
C) Sunlight
D) Wind
Answer: A) Suspended solids

15. Disease-causing microorganisms present in polluted water are called:
A) Pathogens
B) Nutrients
C) Minerals
D) Producers
Answer: A) Pathogens

16. Oil spills primarily cause pollution of:
A) Water bodies
B) Mountains
C) Buildings
D) Roads
Answer: A) Water bodies

17. Discharge of heated water into a water body causes:
A) Thermal pollution
B) Noise pollution
C) Light pollution
D) Soil erosion
Answer: A) Thermal pollution

18. A common source of thermal water pollution is:
A) Power plants and industries
B) Forests
C) Wind farms
D) Rainwater harvesting structures
Answer: A) Power plants and industries

19. Groundwater may be polluted by:
A) Leaching of contaminants
B) Solar radiation
C) Wind movement
D) Sound waves
Answer: A) Leaching of contaminants

20. Pesticides reaching rivers through agricultural runoff cause:
A) Water pollution
B) Water conservation
C) Groundwater recharge
D) Afforestation
Answer: A) Water pollution

21. Contaminated drinking water can cause:
A) Water-borne diseases
B) Hearing loss
C) Noise pollution
D) Wind erosion
Answer: A) Water-borne diseases

22. Which disease can spread through contaminated water?
A) Cholera
B) Diabetes
C) Hypertension
D) Hearing impairment
Answer: A) Cholera

23. Typhoid is commonly associated with:
A) Contaminated food and water
B) Traffic noise
C) Solar radiation
D) Wind energy
Answer: A) Contaminated food and water

24. Excessive algal growth can eventually result in:
A) Depletion of dissolved oxygen
B) Increased drinking-water quality
C) Elimination of nutrients
D) Increased groundwater recharge
Answer: A) Depletion of dissolved oxygen

25. Biomagnification refers to an increase in the concentration of certain pollutants along the:
A) Food chain
B) Water cycle
C) Rock cycle
D) Hydrological cycle
Answer: A) Food chain

26. The accumulation of a substance within an organism over time is called:
A) Bioaccumulation
B) Biodegradation
C) Sedimentation
D) Filtration
Answer: A) Bioaccumulation

27. Untreated sewage should be prevented from entering:
A) Natural water bodies
B) Sewage-treatment plants
C) Collection systems
D) Treatment units
Answer: A) Natural water bodies

28. Domestic wastewater should preferably be treated in a:
A) Sewage treatment plant
B) Wind farm
C) Solar park
D) Thermal power station
Answer: A) Sewage treatment plant

29. Industrial wastewater should be treated before:
A) Discharge into the environment
B) Generation
C) Manufacturing
D) Collection
Answer: A) Discharge into the environment

30. Reducing excessive fertilizer use can help control:
A) Water pollution
B) Noise pollution
C) Light pollution
D) Thermal radiation
Answer: A) Water pollution

31. Proper sanitation helps reduce:
A) Microbial contamination of water
B) Wind speed
C) Solar radiation
D) Atmospheric pressure
Answer: A) Microbial contamination of water

32. Preventing solid-waste dumping into lakes helps:
A) Protect water quality
B) Increase eutrophication
C) Increase turbidity
D) Increase water contamination
Answer: A) Protect water quality

33. Riparian vegetation can help reduce the entry of:
A) Sediment and pollutants into rivers
B) Sunlight into the atmosphere
C) Wind into cities
D) Oxygen into water
Answer: A) Sediment and pollutants into rivers

34. The primary objective of water-pollution control is to:
A) Protect water quality and public health
B) Increase pollutant discharge
C) Increase untreated sewage disposal
D) Increase waste dumping
Answer: A) Protect water quality and public health

35. Noise is commonly defined as:
A) Unwanted or undesirable sound
B) Pleasant music only
C) Visible radiation
D) Water vibration only
Answer: A) Unwanted or undesirable sound

36. The commonly used unit for measuring sound level is:
A) Decibel
B) Joule
C) Newton
D) Pascal
Answer: A) Decibel

37. The abbreviation for decibel is:
A) dB
B) DO
C) BOD
D) COD
Answer: A) dB

38. The instrument used to measure sound level is:
A) Sound level meter
B) Anemometer
C) Barometer
D) pH meter
Answer: A) Sound level meter

39. Which is a common source of urban noise pollution?
A) Road traffic
B) Rainwater harvesting
C) Solar panels
D) Tree plantation
Answer: A) Road traffic

40. Vehicle horns are a source of:
A) Noise pollution
B) Water pollution
C) Soil pollution
D) Thermal pollution
Answer: A) Noise pollution

41. Aircraft operations can contribute to:
A) Noise pollution
B) Groundwater recharge
C) Water conservation
D) Afforestation
Answer: A) Noise pollution

42. Construction equipment can be a source of:
A) Noise pollution
B) Water conservation
C) Solar energy
D) Groundwater recharge
Answer: A) Noise pollution

43. Industrial machinery can generate:
A) Occupational noise
B) Drinking water
C) Solar energy
D) Groundwater
Answer: A) Occupational noise

44. Loudspeakers at excessive volume can contribute to:
A) Community noise
B) Water treatment
C) Soil conservation
D) Air purification
Answer: A) Community noise

45. Continuous exposure to excessive noise can affect:
A) Human health
B) Water hardness only
C) Soil pH only
D) Solar radiation only
Answer: A) Human health

46. Prolonged exposure to high noise levels may cause:
A) Hearing impairment
B) Fluorosis
C) Cholera
D) Typhoid
Answer: A) Hearing impairment

47. Noise pollution can interfere with:
A) Sleep
B) Photosynthesis only
C) Groundwater recharge
D) Water chlorination
Answer: A) Sleep

48. Excessive noise may cause:
A) Stress and annoyance
B) Improved hearing
C) Water purification
D) Increased soil fertility
Answer: A) Stress and annoyance

49. Noise can interfere with:
A) Communication
B) Water filtration
C) Solar-energy production
D) Groundwater recharge
Answer: A) Communication

50. A possible physiological effect of prolonged noise exposure is:
A) Increased blood pressure
B) Fluorosis
C) Cholera
D) Dental caries
Answer: A) Increased blood pressure

51. Noise barriers are used to:
A) Reduce transmission of traffic noise
B) Increase noise levels
C) Treat wastewater
D) Generate wind energy
Answer: A) Reduce transmission of traffic noise

52. Planting vegetation along roads can assist in:
A) Noise attenuation
B) Increasing vehicle noise
C) Increasing horn use
D) Increasing industrial noise
Answer: A) Noise attenuation

53. Proper maintenance of machinery can help:
A) Reduce noise generation
B) Increase noise generation
C) Increase water pollution
D) Increase air pollution
Answer: A) Reduce noise generation

54. Silencers are commonly used to control noise from:
A) Engines and exhaust systems
B) Water reservoirs
C) Solar panels
D) Trees
Answer: A) Engines and exhaust systems

55. Earplugs and earmuffs are examples of:
A) Hearing protection devices
B) Air-pollution monitors
C) Water-treatment devices
D) Solar-energy devices
Answer: A) Hearing protection devices

56. Restricting unnecessary honking is a measure for controlling:
A) Traffic noise
B) Water pollution
C) Soil pollution
D) Thermal pollution
Answer: A) Traffic noise

57. Zoning can reduce noise exposure by:
A) Separating noise-sensitive areas from major noise sources
B) Increasing industrial activity near hospitals
C) Increasing traffic around schools
D) Increasing loudspeaker use
Answer: A) Separating noise-sensitive areas from major noise sources

58. Hospitals and educational institutions are generally considered:
A) Noise-sensitive areas
B) Major industrial noise sources
C) Mining areas
D) Heavy-traffic terminals
Answer: A) Noise-sensitive areas

59. Noise pollution can affect students by reducing:
A) Concentration
B) Water quality
C) Soil fertility
D) Rainfall
Answer: A) Concentration

60. The basic purpose of noise-control measures is to:
A) Reduce harmful noise exposure
B) Increase sound intensity
C) Increase horn usage
D) Increase industrial noise
Answer: A) Reduce harmful noise exposure

61. Air pollution refers to the presence of contaminants in the atmosphere at levels that may:
A) Harm health or the environment
B) Always improve air quality
C) Increase drinking-water quality
D) Improve soil fertility in all cases
Answer: A) Harm health or the environment

62. Pollutants emitted directly from a source are called:
A) Primary pollutants
B) Secondary pollutants
C) Tertiary pollutants
D) Natural resources
Answer: A) Primary pollutants

63. Pollutants formed in the atmosphere through chemical reactions are called:
A) Secondary pollutants
B) Primary pollutants
C) Water pollutants
D) Soil nutrients
Answer: A) Secondary pollutants

64. Which is a major source of urban air pollution?
A) Motor vehicles
B) Rainwater harvesting
C) Tree plantation
D) Solar panels
Answer: A) Motor vehicles

65. Industries can release air pollutants through:
A) Stacks and process emissions
B) Rainwater harvesting systems
C) Groundwater wells
D) Solar collectors
Answer: A) Stacks and process emissions

66. Burning fossil fuels can release:
A) Air pollutants
B) Only pure oxygen
C) Only water
D) No gaseous products
Answer: A) Air pollutants

67. PM in air-pollution studies stands for:
A) Particulate Matter
B) Pollution Measurement
C) Primary Material
D) Pressure Matter
Answer: A) Particulate Matter

68. PM10 refers to particulate matter with aerodynamic diameter generally:
A) 10 µm or less
B) 100 µm or more
C) Exactly 50 mm
D) More than 1 metre
Answer: A) 10 µm or less

69. PM2.5 refers to fine particulate matter with aerodynamic diameter generally:
A) 2.5 µm or less
B) 25 mm or less
C) 250 µm or more
D) Exactly 10 mm
Answer: A) 2.5 µm or less

70. Carbon monoxide is represented by:
A) CO
B) CO_2
C) SO_2
D) NO_2
Answer: A) CO

71. Sulfur dioxide is represented by:
A) SO_2
B) CO
C) O_3
D) CH_4
Answer: A) SO_2

72. Nitrogen dioxide is represented by:
A) NO_2
B) SO_2
C) CO
D) O_2
Answer: A) NO_2

73. Ground-level ozone is represented by:
A) O_3
B) O_2
C) CO_2
D) SO_2
Answer: A) O_3

74. Ground-level ozone is mainly a:
A) Secondary air pollutant
B) Primary pollutant emitted directly by vehicles
C) Water pollutant only
D) Soil nutrient
Answer: A) Secondary air pollutant

75. Ground-level ozone is formed through photochemical reactions involving precursor pollutants in the presence of:
A) Sunlight
B) Darkness only
C) Groundwater
D) Soil moisture only
Answer: A) Sunlight

76. NOx refers mainly to:
A) Nitrogen oxides
B) Sulfur oxides
C) Carbon oxides
D) Hydrogen oxides
Answer: A) Nitrogen oxides

77. VOC stands for:
A) Volatile Organic Compound
B) Variable Oxygen Concentration
C) Vehicle Organic Carbon
D) Volatile Oxygen Compound
Answer: A) Volatile Organic Compound

78. NOx and VOCs are important precursors in the formation of:
A) Ground-level ozone
B) Groundwater
C) Soil minerals
D) Drinking water
Answer: A) Ground-level ozone

79. Incomplete combustion of carbon-containing fuels can produce:
A) Carbon monoxide
B) Oxygen only
C) Helium
D) Neon
Answer: A) Carbon monoxide

80. Road dust is an important source of:
A) Particulate matter
B) Dissolved oxygen
C) Water hardness
D) Fluoride
Answer: A) Particulate matter

81. Fine particulate matter primarily enters the body through the:
A) Respiratory system
B) Digestive system only
C) Skeletal system
D) Hair
Answer: A) Respiratory system

82. Exposure to air pollution can aggravate:
A) Respiratory diseases
B) Water hardness
C) Soil erosion
D) Groundwater recharge
Answer: A) Respiratory diseases

83. Which group may be particularly vulnerable to air pollution?
A) Children and older adults
B) Buildings only
C) Vehicles only
D) Roads only
Answer: A) Children and older adults

84. Carbon monoxide reduces the oxygen-carrying capacity of blood by binding strongly with:
A) Hemoglobin
B) Calcium
C) Glucose
D) Vitamin C
Answer: A) Hemoglobin

85. Ground-level ozone can cause irritation of the:
A) Respiratory system
B) Building foundation
C) Road pavement
D) Water-treatment plant
Answer: A) Respiratory system

86. Sulfur dioxide exposure can particularly affect the:
A) Respiratory system
B) Road network
C) Water distribution system
D) Building foundation
Answer: A) Respiratory system

87. Nitrogen dioxide exposure can irritate the:
A) Airways and lungs
B) Soil only
C) Buildings only
D) Roads only
Answer: A) Airways and lungs

88. Air pollution may reduce visibility due to:
A) Particles and atmospheric pollutants
B) Groundwater recharge
C) Rainwater harvesting
D) Solar panels
Answer: A) Particles and atmospheric pollutants

89. Acid rain is associated mainly with atmospheric emissions of:
A) Sulfur dioxide and nitrogen oxides
B) Oxygen and hydrogen
C) Helium and neon
D) Nitrogen and argon only
Answer: A) Sulfur dioxide and nitrogen oxides

90. Air pollutants can adversely affect:
A) Human health, vegetation and materials
B) Only roads
C) Only buildings
D) Only vehicles
Answer: A) Human health, vegetation and materials

91. Using public transportation can help reduce:
A) Vehicular air emissions
B) Groundwater recharge
C) Water hardness
D) Soil fertility
Answer: A) Vehicular air emissions

92. Regular vehicle maintenance can help:
A) Reduce excessive vehicular emissions
B) Increase air pollution
C) Increase fuel wastage
D) Increase exhaust smoke
Answer: A) Reduce excessive vehicular emissions

93. Cleaner and renewable energy can help reduce dependence on:
A) Fossil fuels
B) Sunlight
C) Wind
D) Solar energy
Answer: A) Fossil fuels

94. An electrostatic precipitator is commonly used to control:
A) Particulate emissions
B) Noise pollution
C) Water hardness
D) Fluoride in water
Answer: A) Particulate emissions

95. A cyclone separator is primarily used for controlling:
A) Particulate matter
B) Dissolved solids in water
C) Traffic noise
D) Water-borne pathogens
Answer: A) Particulate matter

96. A bag filter is commonly used to remove:
A) Particulate matter from gas streams
B) Dissolved fluoride from water
C) Noise from traffic
D) Pathogens from drinking water
Answer: A) Particulate matter from gas streams

97. A wet scrubber can be used to remove:
A) Certain gaseous and particulate pollutants
B) Traffic noise only
C) Groundwater hardness only
D) Solar radiation
Answer: A) Certain gaseous and particulate pollutants

98. Catalytic converters in vehicles help reduce:
A) Harmful exhaust emissions
B) Water pollution
C) Noise from aircraft
D) Soil erosion
Answer: A) Harmful exhaust emissions

99. Avoiding open burning of waste helps reduce:
A) Air-pollutant emissions
B) Groundwater recharge
C) Water conservation
D) Solar radiation
Answer: A) Air-pollutant emissions

100. The overall purpose of air-pollution prevention and control is to:
A) Protect public health and environmental quality
B) Increase pollutant concentrations
C) Increase fuel combustion
D) Increase exposure to pollutants
Answer: A) Protect public health and environmental quality